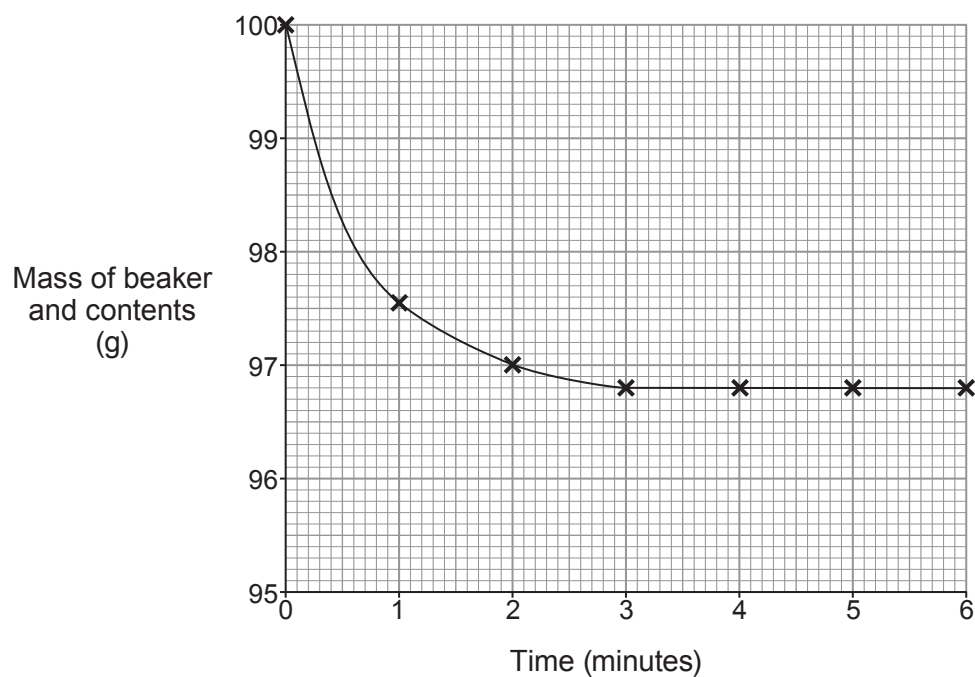


7. Powdered calcium carbonate was added to an excess of dilute hydrochloric acid in a beaker. The mass of the beaker and its contents was recorded every minute for 6 minutes. The graph shows the results.



The equation for the reaction is



- (a) Describe how the mass of the beaker and its contents changes over the first minute. Give the reason for this change. [2]

.....

.....



- (b) (i) Use the graph to determine the time taken for the reaction to finish. [1]

- (ii) When is the reaction at its fastest? Tick (✓) the correct box. [1]

from 0 – 0.5 minutes

☐

from 1 – 1.5 minutes

☐

from 2 – 2.5 minutes

☐

from 3 – 3.5 minutes

☐

- (c) Use your graph to calculate the mean rate of the reaction during the **first two minutes**.
Use the following equation. [2]

$$\text{rate} = \frac{\text{decrease in mass of beaker and contents (g)}}{\text{time (minutes)}}$$

Rate = g/minute

- (d) The experiment was repeated using the same mass of calcium carbonate but as a **lump** instead of a powder.

On the grid opposite, sketch the graph you would expect to obtain from this second experiment. [1]

7

